

Day 5 Assignment – Missing Vital Analysis

Write short paragraphs on any OTHER vital sign or measure that you believe should be in triage but was not included in our dataset. Similar to the last, the main points to include are:

- What the vital sign is?
- What is its clinical range?
- What does it mean for triaging?
- Do add any additional details.

1. Oxygen Saturation (SpO₂)

Oxygen Saturation is a non-invasive vital sign measured via a pulse oximeter that monitors the percentage of haemoglobin molecules in the bloodstream bound to oxygen (Hafen and Sharma 2022). For a healthy adult breathing natural air in a room, the normal clinical range is 95% to 100% (Cleveland Clinic 2026). In emergency triage, it is an indispensable tool for identifying respiratory failure and silent hypoxia, where a patient's respiratory rate (RR) might appear normal but their organs are starved of oxygen. A reading between 91% and 94% signals mild-to-moderate hypoxia/hypoxemia, while any drop to <90% constitutes a medical emergency requiring rapid intervention (Mayo Clinic 2026). The AI triage system will have to treat these threshold values carefully. Patients with existing chronic conditions like COPD usually have a lower, acceptable baseline of 88% to 92%, meaning the AI has to differentiate between a chronic baseline and acute respiratory collapse to avoid false alarms (Cleveland Clinic 2026).

2. Capillary Blood Glucose (CBG)

Capillary Blood Glucose measures the concentration of sugar circulating in a patient's blood. This is typically done by a quick finger-prick test. (Matthew and Zubair 2026; Klonoff, Umpierrez and Rice 2018). Normally, blood sugar ranges between 70 and 140 mg/dL (3.9 to 7.8 mmol/L) (Kearney 2026). For emergency triage, checking blood glucose is mandatory for any patient presenting with an altered mental status, confusion, fainting or a low GCS score. Severely low blood sugar (hypoglycemia, <70 mg/dL) can completely mimic a stroke, seizure or coma but it can be reversed in seconds with immediate glucose administration (Mayo Clinic 2025). Extremely high blood sugar (hyperglycemia, >250 mg/dL) is a sign of a life-threatening diabetic crisis like Diabetic Ketoacidosis (DKA) (Cleveland Clinic 2025). Including glucose data can prevent the AI model from confusing a simple metabolic crash with a severe neurological event.

3. Capillary Refill Time (CRT)

Capillary Refill Time is a quick, non-invasive physical assessment used to check peripheral blood perfusion (Armata et al 2026). It is essentially measured by applying firm pressure to a patient's fingernail bed until it blanches white, then timing how many seconds it takes for the pink colour to return once released. The normal clinical range is <2 seconds for newborns, <3 seconds for adults and slightly >3 seconds for older adults. (Cleveland Clinic 2022). In emergency triage, CRT gives an instantaneous check on a patient's circulatory status and cardiac output. If a patient's CRT is prolonged (>3 seconds), it indicates that the body is constricting peripheral blood vessels to protect core organs. This is an early sign of circulatory shock or severe dehydration. (McGuire, Gotlib and King 2023; Armata et al 2026). This test requires no medical hardware, making it one of the most reliable and easiest triage metrics for resource-constrained environments like those found in the Caribbean.

References

- Armata, Nikol N., Alyssa Haag, Lily Guo, Kelsey LaFayette, and Abbey Richard. 2026. "Capillary Refill Time." Osmosis from ELSEVIER. <https://www.osmosis.org/answers/capillary-refill-time>
- Cleveland Clinic. 2026. "Blood Oxygen Level." <https://my.clevelandclinic.org/health/diagnostics/22447-blood-oxygen-level>
- Cleveland Clinic. 2025. "Hyperglycemia (High Blood Sugar)." <https://my.clevelandclinic.org/health/diseases/9815-hyperglycemia-high-blood-sugar>
- Cleveland Clinic. 2022. "Capillary Refill Time." <https://my.clevelandclinic.org/health/diagnostics/23376-capillary-refill-time>
- Hafen, Brant B., and Sandeep Sharma. 2022. "Oxygen Saturation." StatPearls – NCBI Bookshelf. <https://www.ncbi.nlm.nih.gov/books/NBK525974/>
- Kearney, Mark. 2026. "Blood Sugar." British Heart Foundation. <https://www.bhf.org.uk/informationsupport/heart-matters-magazine/medical/tests/blood-sugar>
- Klonoff, David C., Guillermo E. Umpierrez, and Mark J. Rice. 2018. "A Milestone in Point of Care Capillary Blood Glucose Monitoring of Critically Ill Hospitalized Patients." Journal of Diabetes Science and Technology 12 (6): 1095–1100. <https://pmc.ncbi.nlm.nih.gov/articles/PMC6232725/>
- Matthew, Thomas K., and Muhammad Zubair. 2026. "Blood Glucose Monitoring." StatPearls – NCBI Bookshelf. <https://www.ncbi.nlm.nih.gov/books/NBK555976/>
- Mayo Clinic. 2026. "Hypoxemia." <https://www.mayoclinic.org/symptoms/hypoxemia/basics/definition/sym-20050930>
- Mayo Clinic. 2025. "Diabetic Hypoglycemia – Symptoms & Causes." <https://www.mayoclinic.org/diseases-conditions/diabetic-hypoglycemia/symptoms-causes/syc-20371525>
- McGuire, Duncan, Ari Gotlib, and Jordan King. 2023. "Capillary Refill Time." StatPearls - NCBI Bookshelf. <https://www.ncbi.nlm.nih.gov/books/NBK557753/>